

Wenshuo Wang

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EDUCATION

South China University of Technology, Guangzhou, China

2023.9 - Present

B.Eng, School of Future Technology, Major in Artificial Intelligence

- **GPA:** 3.6/4.0
- **Major Courses:** Fluid Mechanics, Analysis and Applications of Heat, Engineering Math: Calculus II, Engineering Math: Probability & Mathematical Statistics, Discrete Mathematics, Python, Data Structures, Machine Learning, Deep Learning and Computer Vision, Introduction to Cognitive Neuroscience, and related subjects
- **Languages:** English (TOEFL iBT 91)

Massachusetts Institute of Technology, Cambridge, MA, USA

2025.3 - Present

Research Intern (Remote), Kavli Institute for Astrophysics and Space Research (Advisor: Prof. Fan Zhang)

RESEARCH INTERESTS

My primary research goal is to **advance efficient scientific computing through deep learning**. I am particularly interested in: **Spatiotemporal Forecasting** (using deep learning models to iteratively predict spatial data at subsequent time steps, such as flow fields, weather, or traffic flow) and **Scientific Representation Learning** (finding suitable representation forms for physical field data that are appropriate for model learning).

PUBLICATIONS AND PREPRINTS

[1] Derived Fields Preserve Fine-Scale Detail in Budgeted Neural Simulators

Preprint [\[Paper\]](#)

Wenshuo Wang, Fan Zhang[†] ([†] Corresponding Author)

- Propose DerivOpt, a budgeted state-design framework that optimizes which physical fields a neural simulator carries and how storage is allocated across them.
- Reposition carried-state design as the primary bottleneck in budgeted neural simulation by showing that, under tight storage budgets, fine-scale fidelity is often determined when the state is constructed, before stronger architectures, losses, or rollout strategies can recover information already discarded.

[2] Gradient-Informed Temporal Sampling Improves Rollout Accuracy in PDE Surrogate Training

Preprint [\[Paper\]](#)

Wenshuo Wang, Fan Zhang[†]

- Propose GITS, a temporal sampling method that combines pilot-model gradient scores with set-level temporal coverage for PDE surrogate training.
- Challenge the field's default reliance on uniform temporal sampling by showing that training-window selection is itself a core problem for neural PDE surrogates, not one well solved by off-the-shelf samplers.

[3] System-Anchored Knee Estimation for Low-Cost Context Window Selection in PDE Forecasting

Preprint [\[Paper\]](#)

Wenshuo Wang, Fan Zhang[†]

- Propose SAKE, a two-stage low-cost method that uses system-derived anchors and knee-aware search to select context windows for autoregressive neural PDE simulators.
- Shift context-window selection from routine hyperparameter tuning to low-cost algorithmic estimation of near-optimal windows for neural PDE simulators.

[4] Breaking Scale Anchoring: Frequency Representation Learning for Accurate High-Resolution Inference from Low-Resolution Training

ICLR 2026 [\[Paper\]](#)

Wenshuo Wang, Fan Zhang[†]

- Identify Scale Anchoring and propose Frequency Representation Learning for more accurate high-resolution spatiotemporal inference from low-resolution training.
- Correct the field's notion of multi-resolution generalization by showing that stable error across resolutions can mask a deeper failure to learn the true physical operator.

[5] Pi-CCA: Prompt-Invariant CCA Certificates for Replay-Free Vision-Language Continual Learning

ICLR 2026 [\[Paper\]](#)

Jiayu Zhang, Chuangxin Zhao, Ruibo Duan, Canran Xiao, Wenyi Mo, Haoyu Gao, Wenshuo Wang[†]

- Introduce PI-CCA, a replay-free continual multimodal learning method that preserves image-text alignment geometry with compact CCA certificates.
- Enable vision-language models to adapt to non-stationary streams while retaining zero-shot performance and prompt robustness without storing past data.

[6] S²-KD: Semantic-Spectral Knowledge Distillation Spatiotemporal Forecasting

AAAI 2026 [\[Paper\]](#)

Wenshuo Wang, Yaomin Shen, Yingjie Tan, Yihao Chen[†]

- Propose S²-KD, a semantic-spectral distillation framework that transfers causal semantics and spectral knowledge from a multimodal teacher to a vision-only student for spatiotemporal forecasting.
- Recast spatiotemporal distillation as transferring semantic and causal understanding, not just pixel- or frequency-level patterns.

[7] iTAG: Inverse Design for Text Generation with Accurate Causal Graph Annotations

ACL Main 2026 [\[Paper\]](#)

Wenshuo Wang, Boyu Cao, Nan Zhuang, Wei Li[†]

- Propose iTAG, an inverse-design framework that assigns real-world concepts to causal graphs and generates natural text with highly accurate causal annotations.
- Enable scalable benchmarking of text-based causal discovery by providing natural, trustworthy, scalable and low-cost surrogate data where human causal annotations are scarce.

INTERNSHIP EXPERIENCE

HUAWEI Beijing Research Institute, Beijing, China

2024.07-2024.10

Algorithm Engineer Intern

- Developed NLP models for user review analysis and medical text understanding, including text vectorization with XGBoost, BERT-based multi-label classification, T5-based summarization, and evaluation of generative models.
- Designed and implemented a medical conversational agent using Streamlit, SimCSE + Faiss retrieval, Alibaba Cloud text embeddings, and GLM-4-0520 for domain-specific question answering.

COMPETITIONS AND AWARDS

- **Kaggle, Yale/UNC-CH - Geophysical Waveform Inversion.** *Bronze Medal, 81/1365, Top 5.9% (2025)*
- **Kaggle, NeurIPS - Ariel Data Challenge.** *Silver Medal, 49/1151, Top 4.3% (2024)*
- **Kaggle, Home Credit - Credit Risk Model Stability.** *Silver Medal, 23/3856, Top 0.6% (2024)*